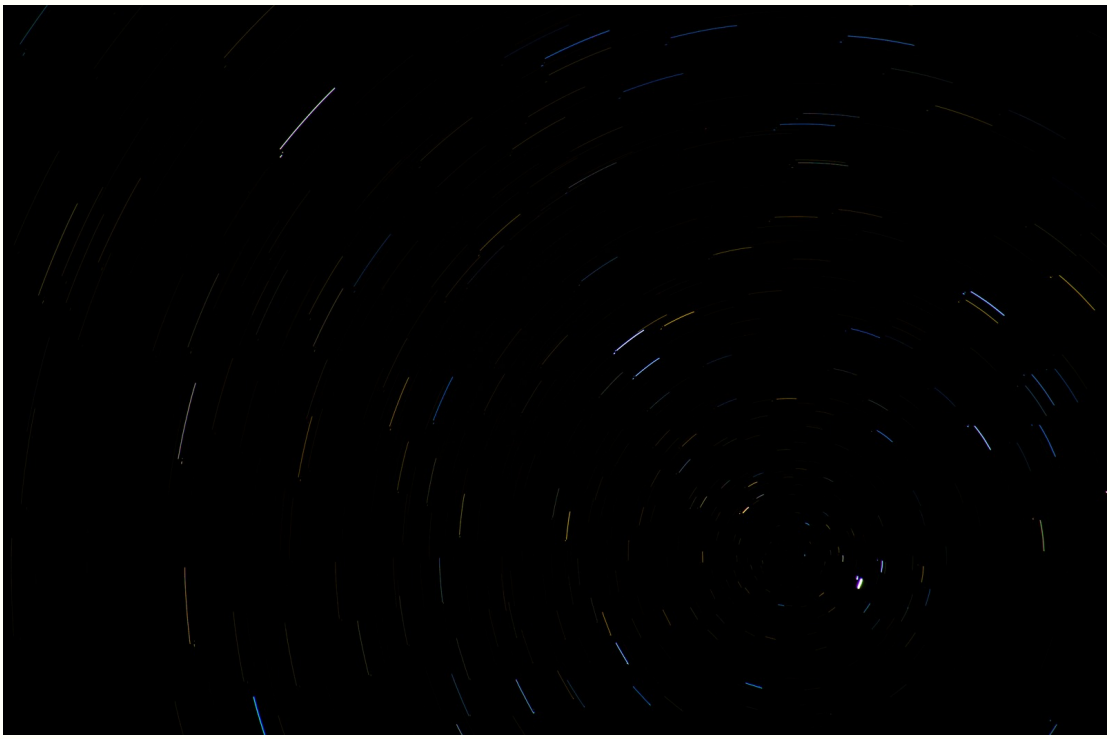
Polaris: the North Star, explained

A Faria lesson for Jeff Franzen · 6 November 2026 (America/New_York)

The whole Wikipedia entry on Polaris, made clear: a triple star, a Cepheid yardstick, the precession of the pole, every name from cynosure to Dhruva, and Shakespeare's Sonnet 116 line by line. Live page: faria.jbf.com/topics/48-polaris/



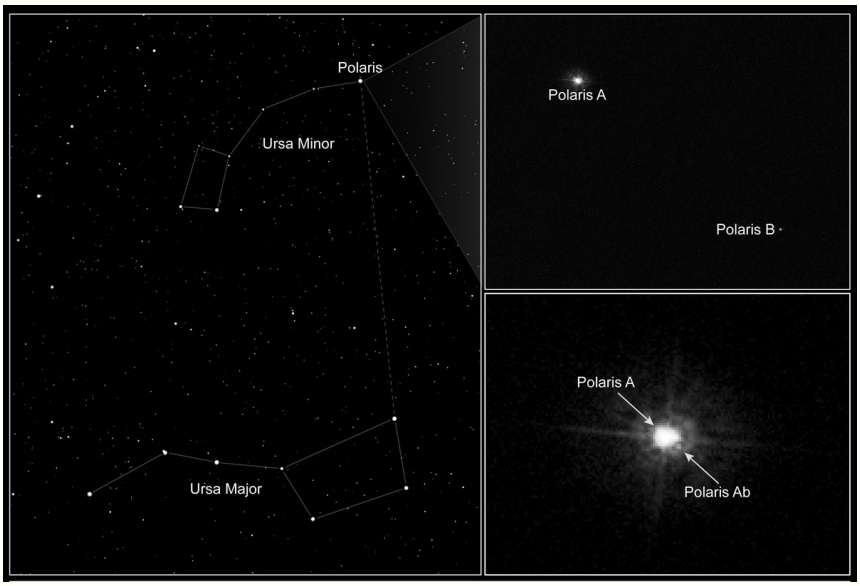
Circumpolar star trails; Polaris near the still center. Daniel Seßer, CC BY-SA 3.0, Wikimedia Commons.

1. Not one star: a multiple system

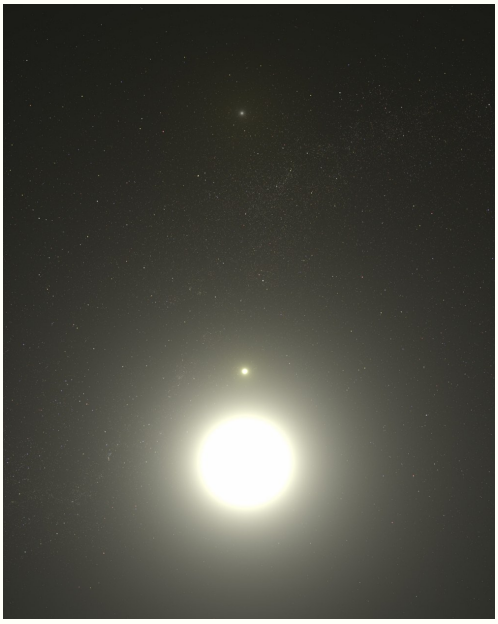
To the naked eye Polaris is a single point of light. It is not. α Ursae Minoris (Alpha Ursae Minoris, α UMi) means "the brightest star of Ursa Minor," the Little Bear. Greek letters were assigned by Johann Bayer in 1603.

- Polaris Aa — yellow supergiant, spectral type F7Ib, mass about 5.13 ± 0.28 solar masses (Evans et al. 2024), radius ~46 solar radii. The star you see; a classical Cepheid.
- Polaris Ab — close dwarf (~F6V), orbits Aa in ~29.4 years. Hubble imaged it in January 2006. Closest approach ~6 AU at the Gaia distance.
- Polaris B — F3V dwarf at ~2,400 AU. William Herschel found it in August 1779 with a telescope he built himself.
- Polaris C and D were once listed as companions but are not physically associated with the system.

Spectral classes F7 / F3. The sequence O B A F G K M runs hot to cool. F stars are yellow-white, a little hotter than our G-type Sun. Ib = luminous supergiant; V = main-sequence dwarf.



Hubble image of the Polaris system. NASA/ESA/HST, public domain.



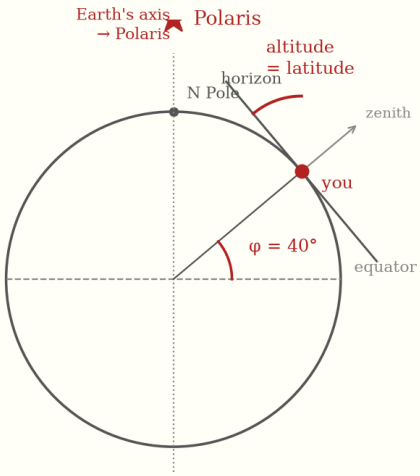
Artist's concept of Aa, Ab, and B. NASA/ESA/G. Bacon (STScI), public domain.

2. Why it stands still

Earth's spin axis points nearly at Polaris, so northern stars wheel around it while Polaris barely moves. In 2018 it was about 0.66° (~40′, about 1.4 Moon-widths) from the true pole, tracing a tiny daily circle ~1.3° across.

The Big Dipper pointers Dubhe and Merak aim at Polaris. Nathaniel Bowditch's American Practical Navigator (1802) lists Polaris among the navigational stars and teaches the clock-face correction from the Dipper.

Altitude of Polaris = latitude. At the North Pole it is overhead; at the equator on the horizon; in Florida (~28°N) about 28° up. That is why latitude was easy for sailors while longitude needed a clock (see the Galileo lesson on faria.jbf.com).



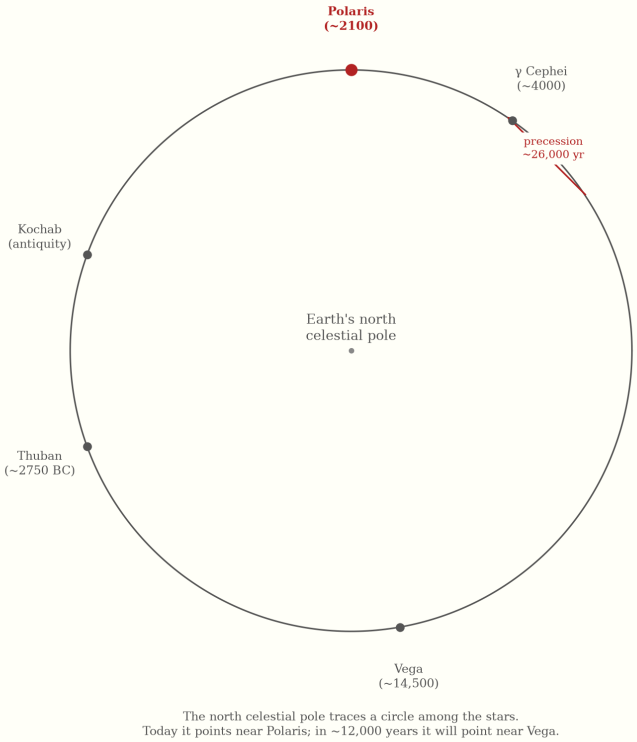
Because Polaris lies almost on Earth's axis, its height above your northern horizon equals your latitude. At the North Pole it is overhead; at the equator it sits on the horizon; in Florida (~28°N) it stands ~28° up.

Altitude of Polaris equals the observer's latitude. Diagram for this lesson.

3. Precession: it will not always be the pole star

Earth's axis wobbles once every ~26,000 years (precession of the equinoxes). The north celestial pole drifts among the stars:

- ~2750 BC — Thuban (α Draconis), near the pole as the pyramids rose
- Classical antiquity — closer to Kochab (β UMi) than to Polaris (~10° from either)
- ~320 BC — Pytheas described the pole as devoid of stars
- 5th c. AD — Stobaeus: ἀεί φανής, "always visible"
- ~2100 AD — Polaris closest (~0.45°)
- ~4000 — Gamma Cephei (Errai); ~14,500 — near Vega; then back after one full cycle

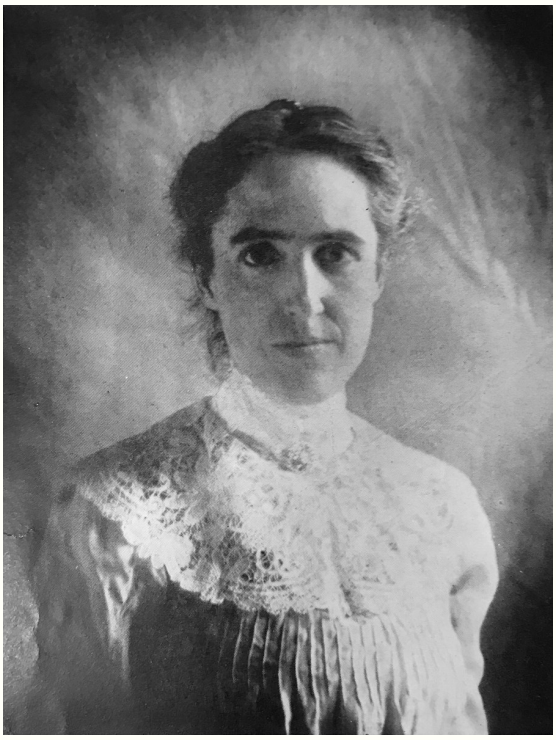


The ~26,000-year precession circle of pole stars. Diagram for this lesson.

4. Cepheids: a pulsing yardstick

Pronounce it SEF-ee-id. A Cepheid swells and shrinks on a steady beat. Named for Delta Cephei.

Henrietta Swan Leavitt (1868–1921), a Harvard "computer," found in 1912 that a Cepheid's period tells its true brightness — the slower the pulse, the more luminous the star. Compare true and apparent brightness and you know the distance. Edwin Hubble used her law on Andromeda Cepheids to show that Andromeda is a separate galaxy and that the universe is expanding.



Henrietta Swan Leavitt. Public domain, Wikimedia Commons.

Polaris is the nearest classical Cepheid (period ~3.97 days). Its amplitude shrank sharply after 1966; the period lengthened ~4.5 s/yr for much of the modern record (Torres 2023 finds a reversal since ~2010). Research in Science (2004) suggests it may be ~2.5× brighter than in Ptolemy's day; Edward Guinan: if real, "100 times larger" than theory predicts.

5. Distance and parallax

Finger-blink analogy: hold a finger at arm's length and blink each eye; it jumps against the background. Astronomers do the same with Earth's orbit (January vs July). That angle is stellar parallax.

- Hipparcos (1989–93, revised 2007): ~432 light-years
- Turner et al. spectral/photometric "short" distance: ~323 light-years
- Gaia DR3 (via companion B): ~446.5 light-years — current best geometric estimate

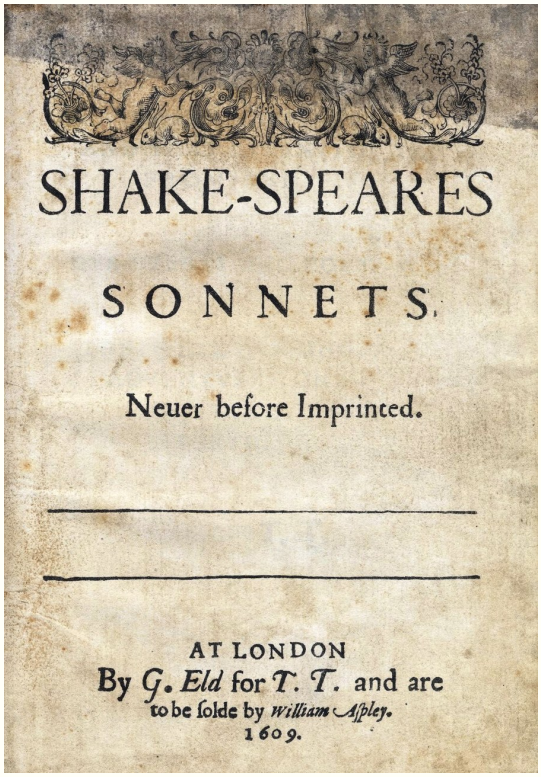
Close companion Ab tugs Aa on a 29-year orbit, mixing orbital wobble with parallax and Cepheid pulsation — binaries spoil clean measurements. Polaris calibrates the cosmic distance ladder; an error here feeds every rung above (galaxies, supernovae, Hubble's constant).

6. Names

- Cynosure (SY-nuh-shoor) — Greek kynosoura, "dog's tail"; thence "center of attention."
- Lodestar — guiding star; kin to lodestone.
- Spenser's "steadfast star" (Faerie Queene I.ii.1): "the northern wagoner had set / His sevenfold team behind the steadfast star."
- Stella polaris / Polaris — Neo-Latin; Alfonsine Tables 1492; IAU proper name 2016.
- Stella Maris / Ave Maris Stella — "star of the sea," Marian title.
- Sanskrit Dhruva — "fixed, constant, immovable."
- Arabic al-Jady ("the kid"); also al-qutb al-shamali ("northern axle") and mismār ("needle").

• Old English scip-steorra ("ship-star"); Rune Poem Tir: "ever on its course over the mists of night and never fails."

7. Shakespeare, Sonnet 116 — the centerpiece


SHAKE-SPEARES

SONNETS.

Neuer before Imprinted.

AT LONDON
By G. Eld for T. T. and are
to be solde by William Aspley.
1609.

Title page, Shake-speares Sonnets, 1609 Quarto. Public domain.

Full text (public domain; 1609 Quarto, modernized spelling):

> *Let me not to the marriage of true minds*
> *Admit impediments. Love is not love*
> *Which alters when it alteration finds,*
> *Or bends with the remover to remove:*
> *O, no! it is an ever-fixed mark*
> *That looks on tempests and is never shaken;*
> *It is the star to every wandering bark,*
> *Whose worth's unknown, although his height be taken.*
> *Love's not Time's fool, though rosy lips and cheeks*
> *Within his bending sickle's compass come;*
> *Love alters not with his brief hours and weeks,*
> *But bears it out even to the edge of doom.*
> *If this be error and upon me proved,*
> *I never writ, nor no man ever loved.*

Close reading of the GIS couplet

> *"It is the star to every wandering bark, / Whose worth's unknown, although his height be taken."*

- Bark = ship (Old French barque).
- The star = Polaris, the sailor's guide.
- Height be taken = measuring the star's altitude with astrolabe, quadrant, or cross-staff — which is the ship's latitude. Exactly the GIS Month rule: altitude of Polaris = latitude.
- Worth's unknown = you can measure and depend on the guide without knowing its true nature. In 1609 no one knew Polaris was a triple Cepheid ~446 light-years away. Love, likewise, can be relied on without being fathomed.

"Ever-fixed mark" = a sea-mark / beacon that looks on tempests and is never shaken — the same steadfastness Spenser gave the pole star. "Edge of doom" = Judgment Day. The closing couplet stakes Shakespeare's entire life's work on the claim.

Pronunciation: Polaris (puh-LAIR-iss); Cepheid (SEF-ee-id); cynosure (SY-nuh-shoor); precession (pree-SESH-un); parallax (PAIR-uh-lacks).

8. Coordinates: sky latitude and longitude

Declination +89° 15′ 50.8″ (sky latitude; +90° = north celestial pole). Right ascension 02h 31m 49.09s (sky longitude, in hours because the sky turns 15° per hour — the same arithmetic as terrestrial longitude).

Review questions

1. At the North Pole, where is Polaris, and what latitude does that report?
2. What did Henrietta Leavitt discover in 1912, and how did that let Hubble prove Andromeda is a separate galaxy?
3. In Sonnet 116, what is a "wandering bark," and what does it mean to "take the height" of the star?
4. Why will Polaris not be the North Star in the year 14,500?
5. Why does a wrong distance to Polaris ripple into measurements of the whole universe?

Answers

1. Straight overhead (zenith); latitude 90°N.
2. A Cepheid's period reveals its true brightness → distance. Hubble found Cepheids in Andromeda far outside the Milky Way.
3. Bark = ship. Taking the height = measuring altitude for latitude.
4. Precession will point Earth's axis near Vega.
5. Polaris is the nearest classical Cepheid; Cepheids calibrate the cosmic distance ladder.

Sources checked against Wikipedia "Polaris" (retrieved 2026-11-06), the 1609 Quarto text of Sonnet 116 (public domain), Spenser Faerie Queene I.ii.1, and the Old English Rune Poem (Tir). Live lesson: https://faria.jbf.com/topics/48-polaris/